

Tactical Surveillance Command Center & Geospatial HUD Architecture

Document Reference: SIH-TR-2026-GUI-02 · Human-Machine Interface (HMI)

1. System Purpose & Operational Overview

The Tactical Surveillance Command Center (gui_app.py) provides the operator interface for transit fleets and municipal control rooms. Built on PyQt5 and QtWebEngine, the application seamlessly fuses live video playback, neural model telemetry, and interactive WebGL cartographic mapping without frame stutter or UI lockups.

Architectural Principle: Decoupled multithreading via PyQt5 QtSignals ensures that heavy PyTorch inference loops and WebGL map rendering never block the main Qt event loop, maintaining a constant **60 FPS UI response**.

2. GUI Subsystem Architecture & Technology Matrix

Subsystem	Technology	Implementation Detail
Desktop Framework	PyQt5 (Qt 5.15.10)	Hardware-accelerated widget hierarchy, high-contrast dark cockpit theme (#0f172a / #1e293b).
Interactive Mapping	Mapbox GL JS WebGL	Embedded via QtWebEngine; renders 60 FPS vector tiles, 3D terrain, and defect pins.
Reverse Geocoding	Mapbox Platform API	Asynchronous thread resolving lat/lon coordinates into street names and school zones.
Telemetry Streaming	Qt Signals & Slots	Thread-safe communication passing video frames, GPS coordinates, and incident structs.

3. Specialized HUD & Geospatial Capabilities

A. 4-Layer Cartographic Map Engine

- **Streets Navigation:** Standard vector arterial view with lane indicators and road classifications.
- **Satellite Imagery:** High-resolution orbital photography for physical road verification.
- **Dark Tactical Nav:** High-contrast nighttime mode with illuminated hazard markers.
- **Polar Radar HUD:** Synthetic tactical layer emphasizing pedestrian crosswalk danger zones.

B. Turn-by-Turn Maneuver HUD (ManeuverHUDWidget)

- Dynamically computes bus speed, corridor speed limits, and pedestrian hazard alerts.
- Visual school zone warning beacon flashing amber within 150 meters of educational institutions.

C. Spatial Pothole Deduplication Algorithm (src/spatial_dedup.py)

- Resolves the multi-frame detection problem using the Haversine formula with a 15-meter geohash radius.
- Merges hundreds of redundant camera frame sightings into single, unique physical road defects.

D. 4-Stage Forensic Incident Dossier Modal (EvidenceInspectorDialog)

- Allows operators to inspect: (1) Raw camera capture, (2) Preprocessed Otsu crop, (3) Bounding box overlay, and (4) Geospatial context with direct Google Maps link.